TENSTAR ROBOT 0.96 Inch OLED SSD1306/SSD1315 128X64 IIC I2C Serial Display Module

TENSTAR 0.96 Inch OLED SSD1306/SSD1315 128x64 IIC I2C Serial Display Module User Manual

Model: 0.96 Inch OLED SSD1306/SSD1315 128X64 IIC I2C Serial Display Module

Brand: TENSTAR ROBOT

1. Introduction

This manual provides detailed instructions for setting up and operating your TENSTAR 0.96 Inch OLED SSD1306/SSD1315 128x64 IIC I2C Serial Display

Module. This compact display is designed for various embedded projects, offering clear visuals with ultra-low power consumption.

The module is compatible with a wide range of control chips, including Arduino, 51 Series, MSP430 Series, and STM32/2. It features an embedded driver/controller and utilizes the IIC (I2C) interface for easy integration.

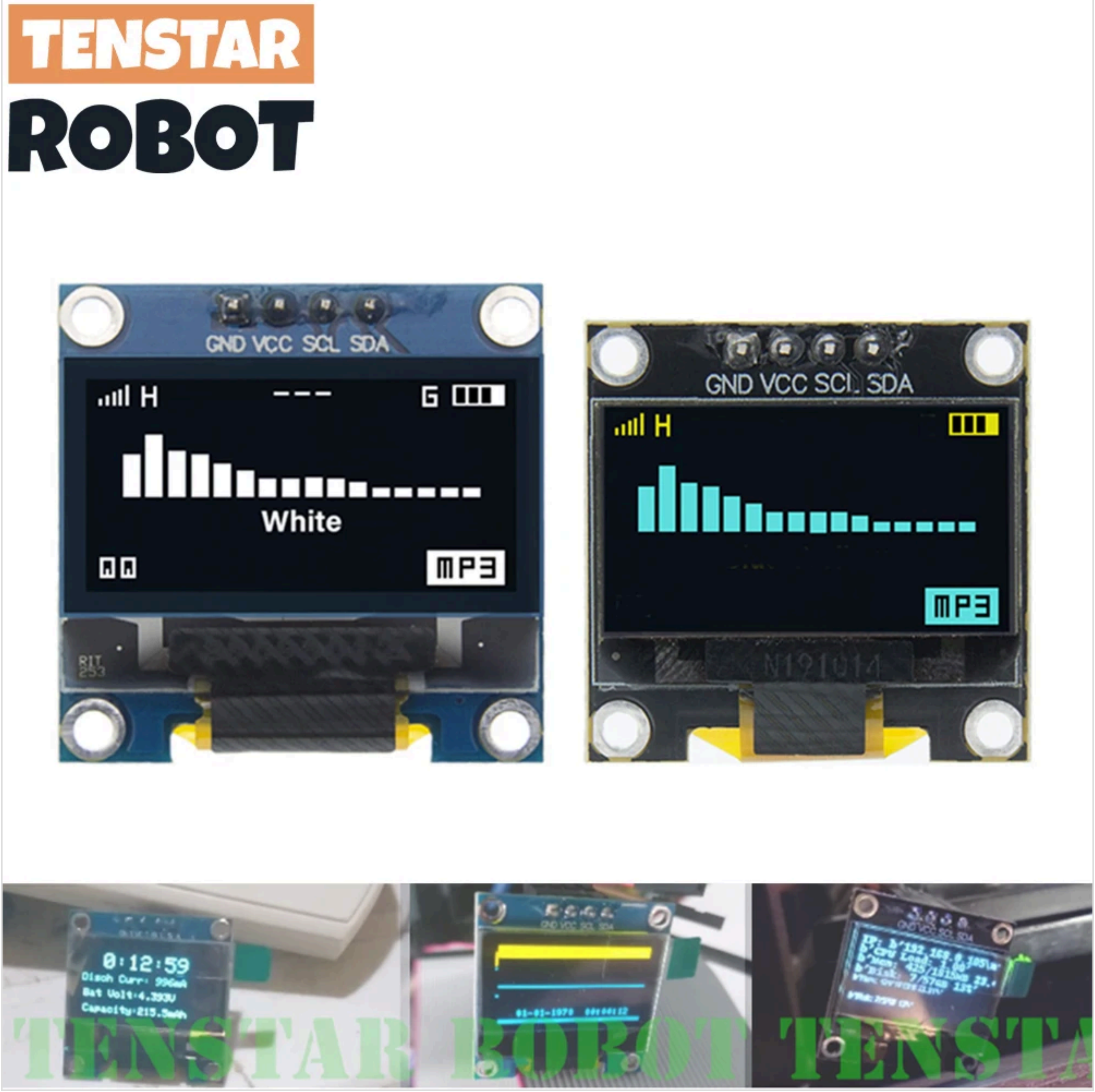

Figure 1: TENSTAR 0.96 Inch OLED Display Modules (White and Blue/Yellow)

2. Specifications

Feature	Specification
Display Type	OLED
Screen Size	0.96 Inch
Resolution	128 x 64 pixels
Interface Type	IIC (I2C) Serial
Pin Definition	GND, VCC, SCL, SDA (4 pins)
Operating Voltage	3V ~ 5V DC
Working Temperature	-30 °C ~ 70 °C
Drive Duty	1/64 Duty
Power Consumption	0.08W (full screen lit)
Panel Dimensions (approx)	26.70 x 19.26 x 1.85 mm
Active Area (approx)	21.74 x 11.2 mm
Driver IC	SSD1306 (Blue board), SSD1315 (Black board)

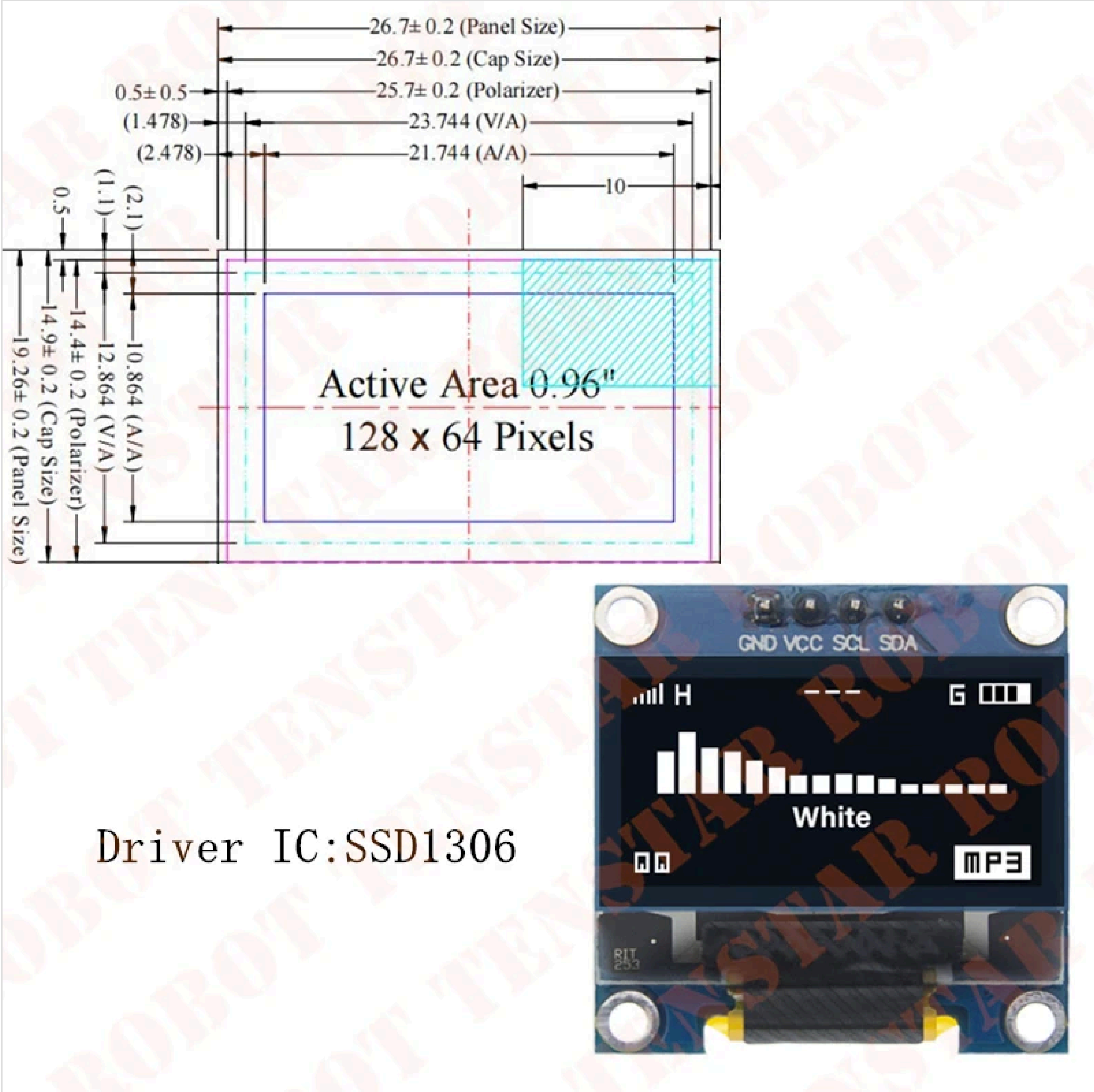

Figure 2: SSD1306 Module Dimensions

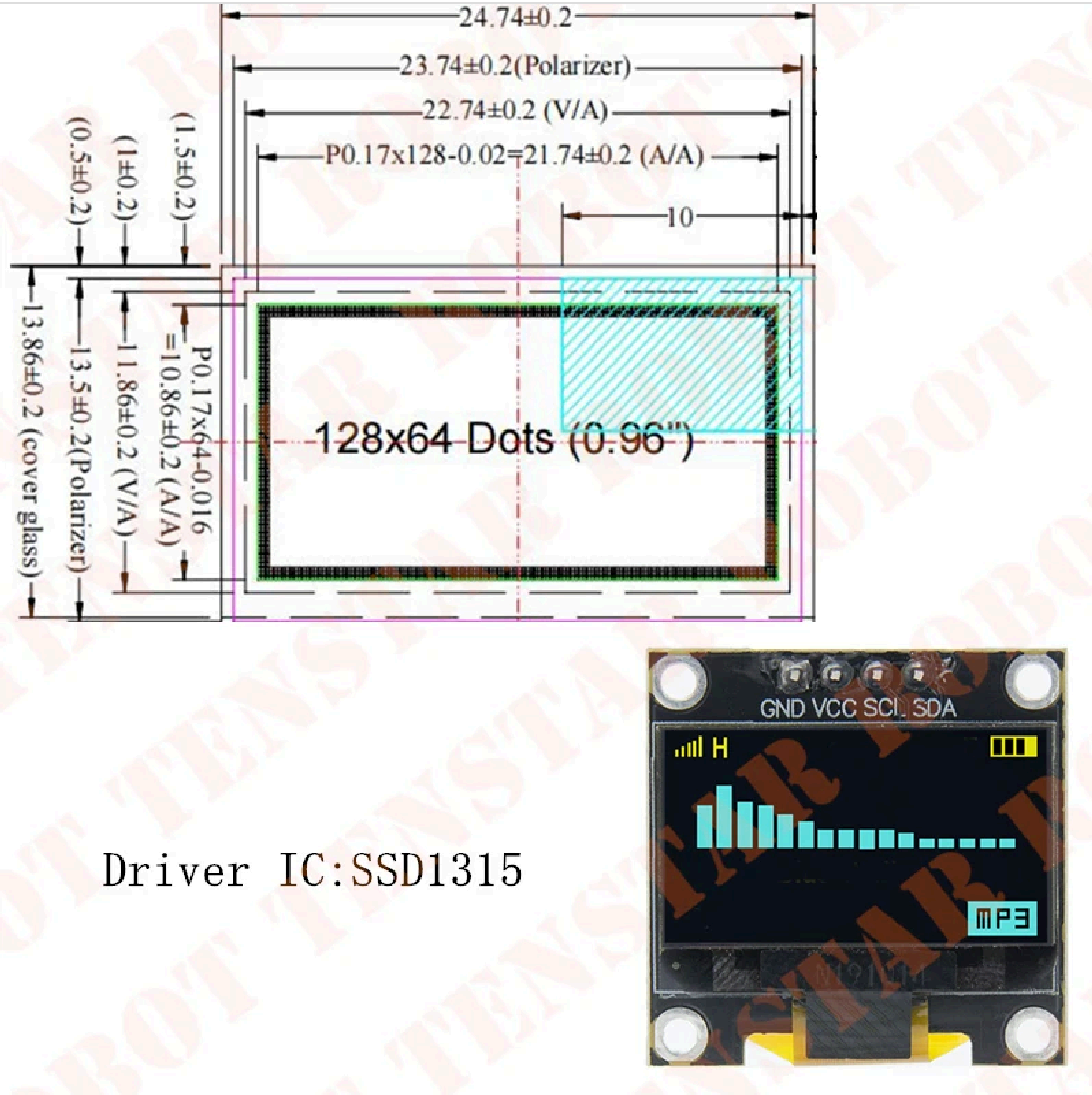

Figure 3: SSD1315 Module Dimensions

Figure 4: Available Display Color Options

3. Setup

3.1 Pinout

The OLED display module features a 4-pin IIC interface:

- **GND:** Ground connection.
- **VCC:** Power supply (3V to 5V DC).
- **SCL:** Serial Clock line for I2C communication.
- **SDA:** Serial Data line for I2C communication.

3.2 Wiring Instructions

Connect the module to your microcontroller (e.g., Arduino) as follows:

1. Connect the **GND** pin of the OLED module to the **GND** pin of your microcontroller.
2. Connect the **VCC** pin of the OLED module to the **3.3V** or **5V** pin of your microcontroller (ensure compatibility with your specific module and microcontroller).
3. Connect the **SCL** pin of the OLED module to the **SCL (A5 on Arduino Uno)** pin of your microcontroller.
4. Connect the **SDA** pin of the OLED module to the **SDA (A4 on Arduino Uno)** pin of your microcontroller.

3.3 IIC Address Selection

The OLED module typically uses an I2C address of 0x3C or 0x3D (7-bit addresses, which correspond to 0x78 or 0x7A in 8-bit format). Some modules provide options to change this address, which is useful when connecting multiple OLED displays to a single I2C bus.

Refer to the images below for common address selection methods:

Figure 5: IIC Address Selection on Blue Board (e.g., by bridging specific pads for 0x78 or 0x7A)

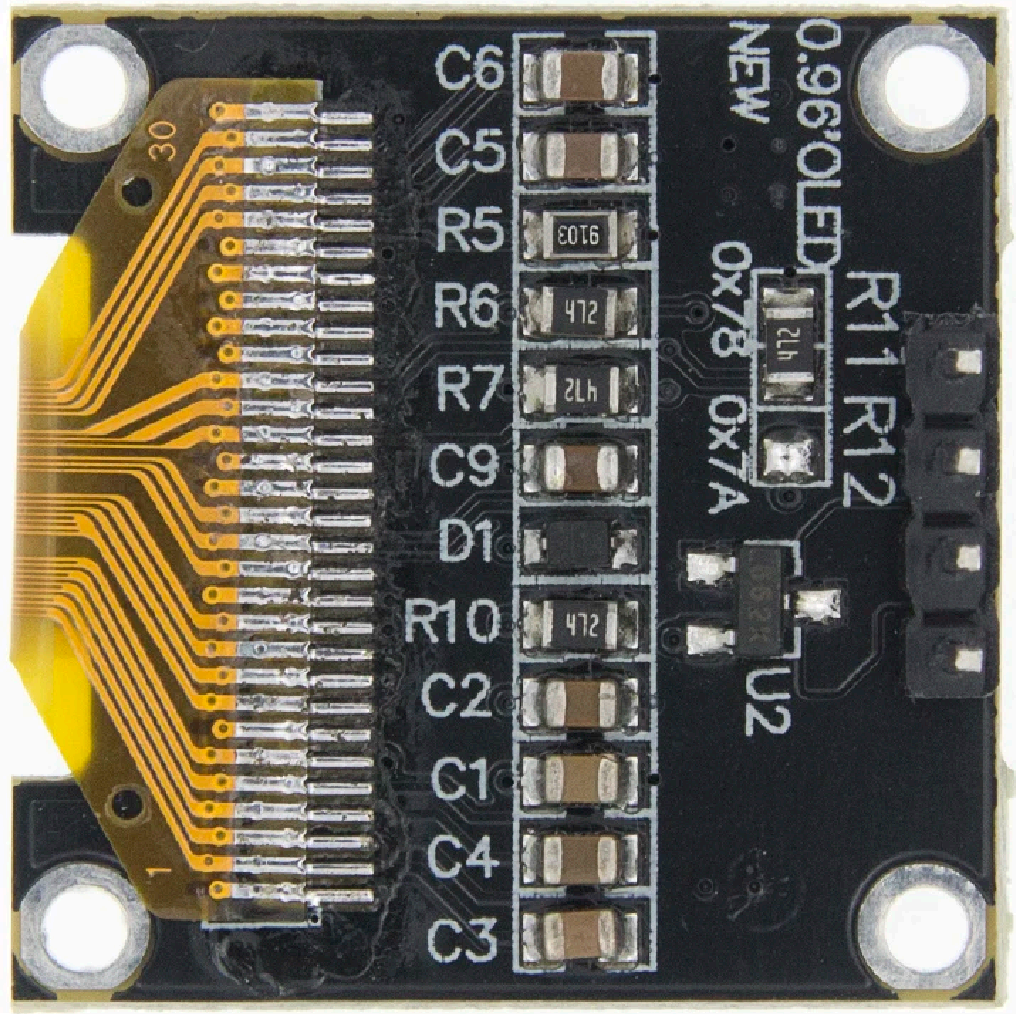

Figure 6: IIC Address Selection on Black Board (e.g., by modifying resistors R11/R12)

Consult the specific module's silkscreen or datasheet for exact instructions on how to change the I2C address, usually involving soldering or desoldering small resistors/jumpers.

4. Operating the Module

4.1 Software Libraries

To operate the OLED display, you will typically need a compatible software library for your chosen microcontroller platform. For Arduino, popular libraries include:

- **Adafruit SSD1306 Library:** Widely used for SSD1306-based OLED displays.
- **Adafruit GFX Library:** A core graphics library required by the SSD1306 library.

4.2 Basic Code Example (Arduino)

After installing the necessary libraries, you can use a basic sketch to test the display. Here's a conceptual outline:

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128 // OLED display width, in pixels
#define SCREEN_HEIGHT 64 // OLED display height, in pixels

// Declaration for an SSD1306 display connected to I2C (SDA, SCL
#define OLED_RESET -1 // Reset pin # (or -1 if sharing Arduino r
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLE

void setup() {
  Serial.begin(9600);

  // SSD1306_SWITCHCAPVCC = generate display voltage from 3.3V i
  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) { // Address 0x
    Serial.println(F("SSD1306 allocation failed"));
    for(;;); // Don't proceed, loop forever
  }

  // Clear the display buffer.
  display.clearDisplay();

  // Display Text
  display.setTextSize(1);
  display.setTextColor(SSD1306_WHITE);
```

```
  display.setCursor(0,0);
  display.println("Hello, World!");
  display.display();
}

void loop() {
  // Your main code here
}
```

Remember to adjust the I2C address (e.g., `0x3C` or `0x3D`) in the `display.begin()` function according to your module's configuration.

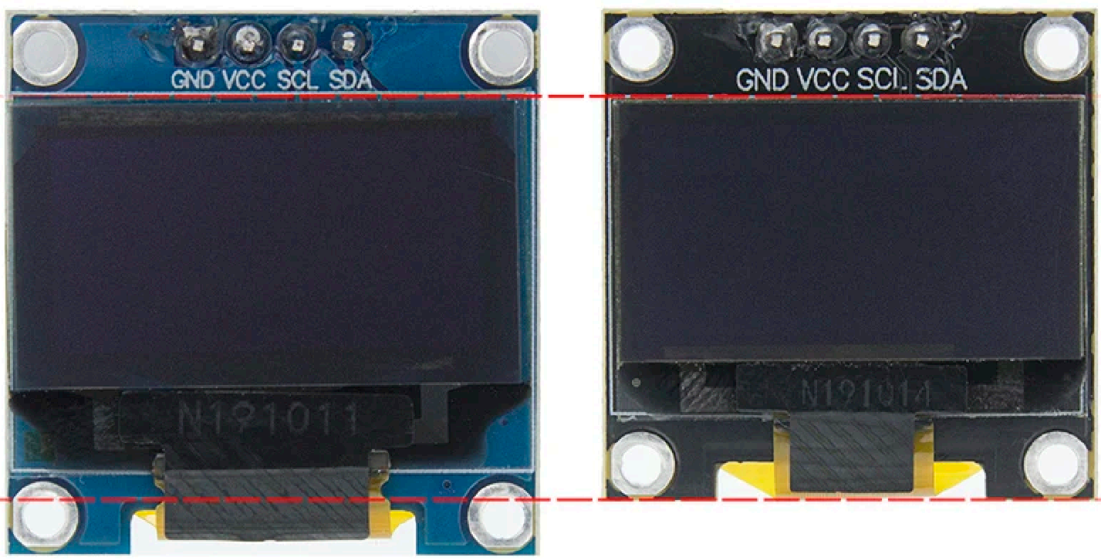

Figure 7: Examples of OLED Module Applications

5. Maintenance

- **Cleaning:** Use a soft, dry cloth to gently wipe the display surface. Avoid using liquids or abrasive materials.
- **Handling:** Handle the module by its edges to prevent damage to the display area or electronic components.
- **Storage:** Store the module in a dry, anti-static environment away from direct sunlight and extreme temperatures.
- **Power Supply:** Always ensure the correct voltage (3V-5V DC) is supplied to prevent damage.

6. Troubleshooting

6.1 Display Not Lighting Up

- **Power Check:** Verify that VCC and GND connections are correct and receiving the appropriate voltage (3V-5V).
- **Wiring:** Double-check all I2C connections (SDA, SCL) for continuity and correct pin assignment.
- **I2C Address:** Ensure the I2C address in your code matches the address configured on your module (typically 0x3C or 0x3D). An incorrect address is a common cause of communication failure.
- **Code Initialization:** Confirm that the display library is correctly initialized in your code.

6.2 Garbled or Incorrect Display

- **Library Compatibility:** Ensure you are using a library compatible with your specific driver IC (SSD1306 or SSD1315) and display resolution (128x64).
- **Buffer Issues:** Make sure you are clearing the display buffer (`display.clearDisplay()`) before drawing new content and calling `display.display()` to update the screen.
- **I2C Communication:** Check for any interference or loose connections on the I2C lines.

6.3 Multiple Displays on One I2C Bus

If you are trying to use multiple OLED displays on the same I2C bus and only one is working, it is likely an I2C address conflict. Each device on an I2C bus must have a unique address.

- **Change Address:** As shown in Figures 5 and 6, some modules allow you to change their I2C address by modifying jumpers or resistors. Ensure each display has a distinct address.
- **Software Configuration:** In your code, you will need to initialize each display with its unique I2C address.

7. User Tips

- **I2C Address Management:** When using multiple SSD1306 displays on a single I2C bus, remember to reconfigure the I2C address for each display to avoid conflicts. This often involves soldering or desoldering specific pads or resistors on the module's PCB.
- **Power Supply Stability:** While the module supports 3V-5V, a stable power supply within this range is crucial for optimal performance and longevity.
- **Experiment with Libraries:** If you encounter issues, try different versions of the display library or alternative libraries that support the SSD1306/SSD1315 driver.

8. Support

For further assistance or technical support, please refer to the TENSTAR ROBOT Store or the product page where you purchased this module. Online communities and forums dedicated to Arduino and electronics projects can also be valuable resources for troubleshooting and project ideas.

Ask about setup, troubleshooting, compatibility, parts, safety, or missing instructions. Manuals+ will review the question and use this page's manual context to help answer it.

Name

Anonymous

Email

you@example.com

Question

Example: How do I reset this device or fix this error?

Details

Model number, symptoms, what you tried, or the section of the manual you are using.

Submit question

TENSTAR ROBOT 0.96 INCH OLED SSD1306/SSD1315 128X64 IIC I2C SERIAL DISPLAY MODULE

KEEP EXPLORING MANUALS, GUIDES, AND SUPPORT DOCS.

Search model numbers, browse help resources, and jump straight into the next PDF manual or troubleshooting page without losing your place.

Save manuals, notes, uploads, and questions

Sign in with Google, X, or a passkey to keep private notes on saved manuals, track public uploads, and revisit questions you posted.

Open login options